



I Thread Intelligence System

& MALWARE DETECTION SYSTEM

Real-Time Cybersecurity with Machine Learning



Problem & Objectives

THE PROBLEM



Rising Cyberattacks

Organizations face thousands of attacks daily — DDoS, port scanning, brute force, botnets.



Manual Monitoring Limitations

Human inspection cannot scale to inspect every network packet in real-time.



Traditional Systems Miss Threats

Signature-based detection fails against novel attack patterns and zero-day exploits.

OBJECTIVES

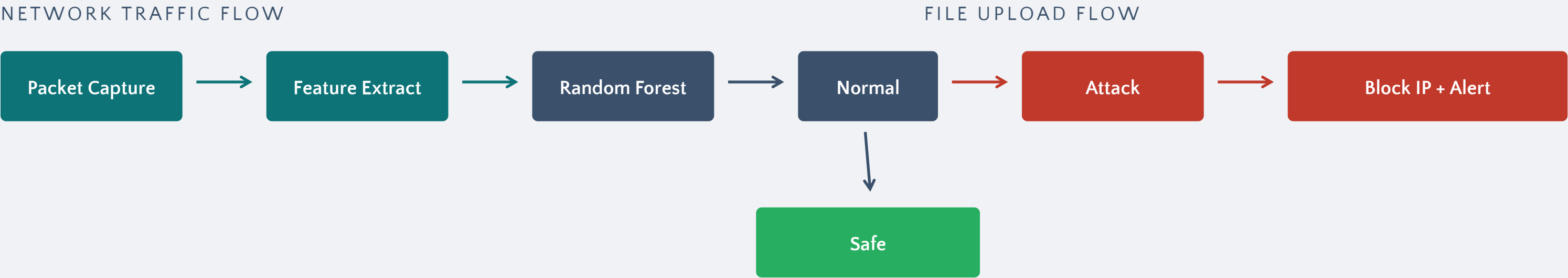
01 Detect Network Intrusions using ML (Random Forest on CIC-IDS2017 dataset)

02 Detect Malware in uploaded files through feature analysis

03 Block Malicious IPs automatically when AI confidence exceeds threshold

04 Real-Time Dashboard for monitoring traffic, alerts, and blocked IPs

How the System Works



KEY COMPONENTS



Flask Backend

REST APIs, SocketIO for live updates, JWT auth, IP blocking middleware



React Frontend

Dashboard with Chart.js, traffic monitor, attack logs, blocked IPs view



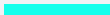
MongoDB

Stores packets, alerts, blocked IPs, predictions, user data, system logs



Docker

Full containerization with Docker Compose for React, Flask, and MongoDB



Technologies & Features

TECH STACK

FRONTEND

React.js • Vite • Tailwind CSS • Chart.js • Axios

BACKEND

Flask • Flask-RESTX • Flask-CORS • Flask-SocketIO

ML & DATA

Scikit-learn • Random Forest • XGBoost • CIC-IDS2017

INFRASTRUCTURE

Scapy • MongoDB • Docker • Docker Compose

KEY FEATURES



Live Traffic Monitoring with real-time graphs and attack timeline



AI Intrusion Detection for 7 attack types with confidence scores



Malware File Scanning with safe/malicious classification



Automatic IP Blocking with configurable thresholds and timers



Role-Based Access with JWT auth (Admin & Viewer roles)



Dark Mode, PDF/CSV Export, and responsive UI design

RESULTS & FUTURE SCOPE

KEY RESULTS

- ✔ Detects attacks in **real-time** with confidence scores and threat levels
- ✔ Identifies **malware** through automated file analysis
- ✔ Improves **incident response time** through automatic IP blocking
- ✔ Enhances security **visibility** via comprehensive dashboard

FUTURE SCOPE

- **Deep Learning** models (LSTM, CNN) for improved accuracy
- **Zero-day attack** detection capabilities
- **Cloud deployment** on AWS or GCP
- **Email and SMS** alert notifications

THANK YOU
